

Chapter 5: History of the Parallel Postulate:

Projects:

1. Cite:

Torretti, R.

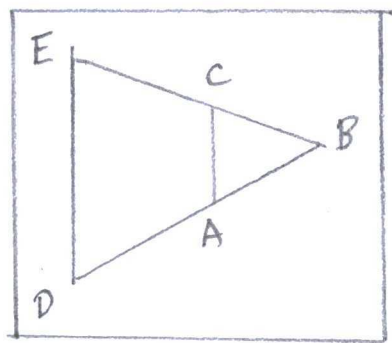
"Philosophy of Geometry from Riemann
to Poincare"

University of Puerto Rico (1984)

The universe had multiple models: bounded and unbounded. Eudoxus (~400 BC) developed a solution to planets from uniform motions with Mercury, Venus, Mars, Jupiter, and Saturn.

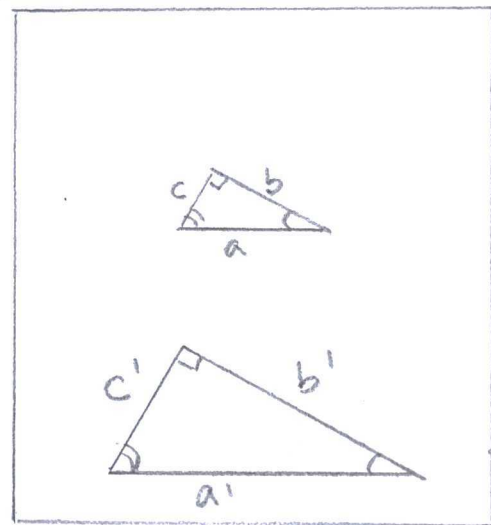
Circular motion suited phenomena in night sky. Eudoxus bounded rotation around many centers. Later Renee Descartes' (~1600) method of coordinates invigorated astronomy. Cartesian coordinates unbounded space around an origin. Timelines indicated two models in astronomy: (i) all spheres rotate around a center, (ii) many centers for rotation.

2. Rene Descartes in "La Géométrie" defined similar triangles through multiples and proportions.



For example, parallels DE and CA were multiples in a triangle. BE and BD related through AC.

Earlier, Euclid provided similar triangles in Books IV and VI. He had exact ratios across side lengths: $a/a' = b/b' = c/c'$.



Similar triangles have a 2000-plus year history.

3. Geometry in one-dimension does not correspond to two-, or three-dimensions. The plane in three-dimensions maps to a line in two-dimensions, then maps to a point in one-dimension. While spheres are equivalent ^{to} one-dimensional points. Other dilemma arise across dimensions, such as number of points at intersection between shapes. Abstract geometry requires logical correspondence between dimensions, fields, or numbers.